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Financing technological innovation in China: neo-developmental financial statecraft through government guidance funds

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ABSTRACT

This paper examines China's distinctive approach to financing innovation, contributing to debates on the derisking state, neo-developmental alternatives, and economic statecraft. It focuses on the central role of government guidance funds (GGFs)—jointly owned by a wide range of state entities—and their divergence from government-owned venture capital models in the European derisking state and liberal-developmental states like South Korea. By advancing a financial statecraft perspective, the paper identifies four core features that set Chinese GGFs apart: clearly defined sectoral innovation targets; the conversion of fiscal resources and SOE assets into risk capital; and the coordination in funding the objectives of government planning apparatuses and their geographically conditioned investment mandates. These features reflect both structural aspects of China's political economy and the rapid institutional evolution of its innovation-oriented bureaucracy. The shadow cases from Europe and South Korea underscore the importance of studying the nexus between national varieties of financial statecraft and national policy regimes.

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Introduction

Driven by geopolitics and the threat of climate change, industrial policy has reclaimed center stage in the comparative political economy of national economies, irrespective of their level of per capita income (Babić & Mertens, 2025; Bulfone, 2023; Thurbon et al., 2021, 2023). However, this turn has

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been criticized for serving primarily to entice private capital into serving public agendas in ways that reduce the state to a ‘derisking’ state (Dafermos et al., 2020; Gabor, 2021, 2023; Gabor & Braun, 2023). The alternative proposed by critics is a more directive (‘developmental’) role for the state, with conditionalities (or ‘disciplines’) ensuring that private capitalists align with the state’s industrial policy objectives (Gabor & Braun, 2023; Lee, 2021; Thurbon et al., 2021, 2023).

This paper intervenes in this debate by focusing on the case of China’s state-owned non-bank financial channels specialized in financing the innovation-focused side of industrial policy: government guidance funds (GGFs). While government venture funds (GVCs) and GGFs aim to promote innovation and industrial upgrading by directing capital into strategic sectors, China’s GGFs go beyond the early-stage financing specific to GVCs. Indeed, they are multi-tiered policy tools that not only fund start-ups but also actively support scaling up, industrial clustering, and national strategic coordination. In a literature dominated by an emphasis on state banks (for overviews, see Ban, 2013 and Thurbon, 2016; Kedward et al., 2024; Larsen, 2022; Naughton & Tsai, 2015 on China; for criticism, see Petry et al., 2023), subsidies (Juhász et al., 2024), credit guidance *via* state-owned banks (Ban, 2013; Mikheeva, 2024; Thurbon, 2016), the political economy canon opened only recently to less conventional, non-bank funding channels, such as sovereign wealth funds, state-owned insurance companies, or state-owned enterprise (SOE) pension funds (e.g. Musacchio & Lazzarini, 2014).

While non-bank funding channels, such as GGFs have garnered significant attention from scholars of China’s political economy (e.g. Beck & Larsen, 2025; Naughton, 2021), they remain underexplored as a subject of comparative analysis. This contrasts sharply with the growing comparative literature on the integration of capital markets into state-capitalist frameworks (Petry et al., 2023). Addressing this gap is crucial, as it shifts the focus to a pivotal but often neglected aspect of state-led finance: its capacity to tackle the most challenging dimension of industrial policy—driving technical innovation (Aiginger & Rodrik, 2020; Chang & Andreoni, 2020; Lee, 2021).

To date, comparativists have addressed this challenge by looking at how states created state venture capital institutions (Mazzucato, 2021) or remade their financial systems to encourage the emergence of Silicon Valley-style venture capital ecosystems (Klingler-Vidra, 2018): select high-growth entrepreneurial ventures, supply risk capital, provide advisory and networks, advice or certify to follow-on lenders or investors the higher potential of the start-up (Mocanu & Thiemann, 2024, p. 437). In contrast, less is known about how GVCs operate differently across the different typologies of developmental states or in comparison with

derisking states. The paucity of in-depth and comparative political economy work on GVCs is striking given that they have been familiar to political economists of Chinese industrial policy as well as economists (see below) as an important arm of innovation policy in many capitalist economies. Institutions like the European Investment Fund (EIF), Innovation Fund Denmark, or the South Korea Venture Investment Corporation have domestic innovation as their primary mandate. Indeed, many single case studies in economics documented the effectiveness of GVCs in ‘bridging the equity gap’ in domestic technological innovation—from Europe (Alperovych et al., 2020), Japan (Kiriata, 2018), to China (see Section GGFs as financial statecraft in this paper for a review).

This paper aims to fill in this gap by looking at China’s GGFs. These are hybrid institutions combining a primary GVC function with secondary functions, such as industrial deepening, cluster formation, or even rescue support for firms with critical technologies. Our aim to study them in a comparative context with a representative neo-developmental state (South Korea) and a representative derisking context (European Union—EU). There is by now clear evidence that China’s GGFs stand out in terms of their extraordinary target capital size (US\$1.78 trillion (tr))¹ and critical role in financing China’s leapfrogging of advanced Western and Asian incumbents in fields, such as electronics and cleantech (Luong et al., 2021; Wei et al., 2023).

GGFs can be understood as the financial backbone of China’s ‘financial statecraft’. Building on Weiss and Thurbon’s (2021) work on economic statecraft, we argue that China’s GGFs, particularly in the realm of innovation policy, represent a distinct sub-variety that we term *financial statecraft*. By this, we mean government-led financial mechanisms designed to promote industrial upgrading through domestic technological innovation, with the strategic aim of fending off, outflanking, or keeping pace with clearly identified economic or geopolitical rivals.

The main argument of the paper is that China’s GGFs are not just different from GVCs in derisking states but also embody an ‘dirigiste’ form of financial neo-developmentalism. Unlike GVCs in derisking states like the EU, China’s GGFs keep the state in the ‘driving seat’ of innovation policy finance (Gabor, 2023) when mobilizing capital for industrial innovation, prioritize technological innovation over profits and coordinate mainly with state-owned rather than privately-owned entities (Luong et al., 2021; Wei et al., 2023). However, the phrase ‘the state is in the driving seat’ should be understood less like seamless top-down imposition or command-and-control by the central government, and more like an adaptive form of dirigisme or ‘grand steerage’ (Naughton, 2021) executed by central, provincial and municipal level authorities (Gomes & ten Brink, 2023; Whitfield, 2023).

We further argue that unlike in representative neo-developmental states, such as South Korea, China's GGFs localize investments competitively and experimentally while prioritizing sector-specific over generalist financing. Most importantly, in so doing GGFs forge intricate synergies between the state as a venture and/or equity investor and the state as a planner, orchestrating industrial upgrading across multiple sectors within the highest form of institutional coordination in political economy: macro-planning (Ban & Hasselbalch, 2025). Otherwise put, Chinese financial statecraft leverages state assets beyond budgetary appropriations and state bank investments through employing the financial instruments of guidance funds to scale up industrial policy in ways that reinforce state guidance of the market. Finally, the econometric evidence on China that we reviewed suggests that, despite their problems, China's GGFs have been a 'good enough' solution to the country's innovation challenges. In their flawed but ultimately effective workings, they resemble China's adaptive, learning-oriented bureaucracy, which is specialized in fostering innovation (Gomes & ten Brink, 2023).

The paper begins with a review of the literature on financing innovation and presents our analytical propositions. Section Methodology and data outlines the methodology and data. Section GGFs as financial statecraft frames GGFs as instruments of financial statecraft, examines their mechanisms, effectiveness, and localized operations. Section Chinese GGFs in comparative perspective compares Chinese GGFs with government venture capital in Europe and South Korea, and the final section concludes.

Theoretical framework

Much of the innovation economics literature (see Bazhal, 2017) has highlighted the importance of fiscal and regulatory derisking—such as subsidies, intellectual property rights, grants, and tax credits—to lower private costs, particularly for high-risk innovation. Other work has emphasized public funding for basic research and university partnerships, including tools like public procurement to support commercialization. Political economists have extended this by showing how states actively shape innovation systems: attracting talent, building clusters, launching mission-driven innovation (Mazzucato, 2021), fostering venture capital ecosystems (Klingler-Vidra, 2018), and integrating these strategies, as in China (Sun & Cao, 2023). Yet even where such measures exist, private venture capital is often scarce, and successful start-ups frequently succumb in the capital-intensive 'valley of death' where commercialization must scale, facing continued investor reluctance (Branscomb & Auerswald, 2003).

For a long time, in derisking states, GVC schemes have served primarily as bridge financing for startups (Mocanu & Thiemann, 2024). This reflects the EU's traditional approach to innovation policy, where the state acted as a minimalist repairer of market failures and insurer against private investment risks. Rather than instruments of financial statecraft, these GVCs functioned largely as stopgap, market-fixing measures. The role of the state, however, need not to be confined to derisking alone, but embedded in the kind of developmental network state arrangements that Di Carlo and Schmitz (2023) have highlighted in the case of the EU. Recent work on the derisking–developmental state spectrum (Gabor, 2023; Mazzucato, 2021; Mocanu & Thiemann, 2024; Weiss & Thurbon, 2021) has suggested that GVCs can operate as instruments of financial statecraft within mission-oriented neo-developmental states. This has been shown in South Korea (Klingler-Vidra & Pardo, 2020; Pardo & Klingler-Vidra, 2019) and, on a smaller, experimental scale, even within the EU's derisking framework (Mocanu & Thiemann, 2024). In China, the role of developmental financial statecraft is even more pronounced: GGFs—acting as GVCs or funding adjacent innovation infrastructure—have underpinned industrial upgrading in strategic sectors, such as AI, cleantech, advanced electronics, and dual-use technologies (Luong et al., 2021; Wei et al., 2023).

Yet in comparative political economy, China's GGFs have only recently gained attention. Most analyses remain rooted in economics or economic geography, with limited engagement from political economists. Kennedy (2024) has referenced them briefly, while Beck and Larsen (2025) have examined their role in China's 'market-driven, government-led' green financial strategy. Naughton (2021) has provided a detailed institutional account but focused mainly on their early development. While these studies are important, the literature lacks a deep analysis of how the mechanics of Chinese GGFs function and a comparative perspective to distinguish them from their counterparts in derisking and neo-developmental states. This limits theory-building and risks exoticizing China's model of financial statecraft.

This paper builds on comparative research on China's financial sector (Petry et al., 2023) to position GGFs as instruments of dirigiste neo-developmentalism, driven by the imperative to catch up with—and eventually surpass—technological rivals (Lee, 2021). Following classical developmental state theory (Amsden, 1989; Wade, 1990), GGFs use both carrots (equity financing) and sticks (conditionalities aligning firms with state goals). However, unlike their classical counterparts, neo-developmental strategies operate in a post-World Trade Organization (WTO) global economy with tighter external constraints, limiting overt mercantilism (Wade, 1990). We draw on Ban's (2013) definition of neo-developmentalism

as a hybrid of liberal economic tools and developmentalist aims, focused on labor mobilization, productivity growth, and directing finance toward high-value-added sectors. Embracing trade openness, it modernizes developmentalism through targeted industrial policy and strategic integration into global markets.

In the following, we raise four propositions. A thin form of derisking state can take effect in driving forward innovation only to the extent that private firms' profit motives align (or can be made to align) with the sector-specific innovation goals defined by the state's vertical industrial policy that innovation policy is part of. Based on these considerations, we expect that *while GVCs in derisking states do not have strict technology innovation mandates for specific sectors, neo-developmental states do*. To do this, we used desk research and then looked at fund sites and VC investment monitoring services to identify whether these mandates exist and are fulfilled.

Some scholars (Ban, 2013; Ban & Hasselbalch, 2025) have distinguished between *liberal* neo-developmentalism, where state ownership in finance and industry is relatively limited and where planning is at best suggestive and sectoral (such as South Korea), from *dirigiste* neo-developmentalism, where planning state ownership is significant and planning is indicative and inter-sectoral, as in postwar France or contemporary China (Ban, 2013; Thurbon, 2016). Thus, for all the vibrancy of the Chinese private sector, state ownership dominates: SOEs control 60% of all assets (Gabriele, 2020) and of the 143 Chinese firms listed in the Fortune 500 list in 2021, 82 were state-owned²; state ownership is significant in sectors like machinery, automotive, electronics, and pharma, while it is nearly total (infrastructural) in oil, petrochemicals, electricity, telecommunications, banking, and the military industry; 96 giant SOEs in these sectors serve as critical leverage points for executing state plans, and high-tech firms either owned by the state (China Telecom, Sinopharm) or very close to it (Huawei) serving as forms of critical high-tech infrastructure for privately-dominated high technology sectors like electronics, electric cars, biotech (Medeiros & Majerowicz, 2022).

Regardless of whether one views China as a distinctive form of capitalism (Naughton & Tsai, 2015) or market socialism with Chinese characteristics (Fraser & Jaeggi, 2018), such a scale of state involvement is unprecedented in any other contemporary neo-developmental state and cannot but impact GGF structure and mandates. Hence, we submit that *given this dominant state ownership in the Chinese economy, on the neo-developmental financial spectrum only China's GGFs can mobilize both fiscal resources and broader state-owned assets (such as SOE resources) into risk capital for technological innovation ventures*. To explore this proposition, we compare at ownership structures in Chinese GGFs, EU GVCs, and South Korean GVCs.

Third, China integrates central and provincial indicative planning with the political objectives of the party-state, at the heart of which is President Xi Jinping's vision of 'national rejuvenation'. Importantly, this rejuvenation is to happen mainly through technology innovation, an objective that the Chinese state sees as having an existential geopolitical dimension (Cheung, 2022).³ As such, since 2015, the government's inter-sectoral planning, whose strongly indicative function is backed by resources ('carrots') from state-owned banks and SOEs, has been geared to recalibrate the economy, shifting investments toward cutting-edge industries—a process recently accelerated by the decline of construction and the increase in the cost of labor. Indeed, for China's highest party-state authorities, this mission is deemed existential in a national security sense: 'It is the nation's destiny to be innovation-driven. The core support of national strength is technological innovation capability. National prosperity follows from strength in innovation, and *national misfortune follows from weakness in innovation*'.⁴

Given the existential importance of technological innovation for state planning, the well-evidenced reflexivity of China's innovation policy bureaucracy in terms of adjusting the plans to private sector capabilities and market dynamics (Gomes & ten Brink, 2023) and the tendency of specific institutions in bureaucratic states to align with larger institutional complexes (DiMaggio & Powell, 1983), we propose that *on the neo-developmental financial statecraft spectrum, Chinese GGFs are forms of dirigiste neo-developmentalism whose mandates are embedded in the deepest forms of coordination with the state*. To explore this, we looked at how aligned are the mandates and their actual execution (*via* sectoral specializations of their funds) with the sectoral priorities of strategic planning documents issued by their principals (five-year plans in China, industrial policy strategies in South Korea, and at the EU level).

China's developmental model is unique in its use of experimental and competitive regionalism—what Heilmann and Perry (2020) have called 'adaptive governance'—where provincial bureaucracies compete to fulfil central objectives set in five-year plans. This structure allows regional policymakers to adjust strategies based on local outcomes while insulating the central government from the fallout of failures (Gomes & ten Brink, 2023; Naughton, 2021). Though this decentralization has at times fostered cronyism, its incidence has declined under Xi Jinping (Bulman & Jaros, 2021; Naughton, 2021), and it has enhanced central oversight by tying local autonomy to financial accountability. Economically, regional SOEs increasingly adopt market-oriented tools—such as financial instruments and shareholder value metrics—reflecting a hybrid logic of dirigiste steering and capitalist rationality (Chen & Rithmire, 2020; Naughton, 2021). In contrast, neither the EU's intergovernmental structure nor that of South Korea have such mechanisms: The Commission (the Ministry) cannot

impose experimental governance or competitive incentives across member states (regions), and the EU may instead coordinate through a looser ‘network state’ model focused on regulatory interventions (Di Carlo & Schmitz, 2023).

Therefore, we propose that *based on the combination of the planned and decentralized-experimental nature of Chinese political economy across sub-national units we captured so far, we suggest that Chinese GGFs have stronger geographic conditionalities on their investment than the GVCs based in the EU and South Korea.*⁵ To capture these dynamics, we looked at relevant regulatory frameworks applicable to them as well as at the linkages between central and local actors in terms of shareholding in local GVCs.

Methodology and data

This paper uses an exploratory single case study (China) approach with comparative shadow cases (EU and South Korea). This approach is used in the political economy to investigate a phenomenon or context that is not well understood and to generate hypotheses or identify variables for further study (e.g. Choung et al., 2014).

The shadow cases are drawn based on their representative character for derisking states (EU) and liberal neo-developmental states (South Korea), respectively. In terms of data, we drew first on primary official documents, press releases, as well as a substantial body of literature in economics and economic geography that to date has not been explored by comparative political economy scholarship. We further explored statistical data collected by Chinese corporate registers accessed *via* QiChaCha (<https://www.qcc.com>) as well as by Zero2IPO, a pay-for-service dataset available from inside China. We also processed data collected from individual GGF websites and international consultancy reports. These databases enabled us to examine more closely the shareholding and sectoral targets of GGFs, offering descriptive statistical depth to the analysis.

Additionally, we complemented with 21 semi-structured interviews. Of these, 16 were conducted in China with government VC management firms, private VC management firms working with GGFs, and experts familiar with Chinese innovation policy. These interviews were conducted in Chinese, ensuring linguistic accuracy and enabling participants to articulate sensitive perspectives candidly. The interviews bridge the gap between the stated objectives of GGFs and the realities of their implementation, providing nuanced insights into the operational mechanics of these government financial vehicles. While their organizational affiliations were disclosed, interviewees’ identities remain anonymized to respect confidentiality. The other five interviews were

carried out at the European Investment Bank (EIB), its principal (European Commission—EC), the EIF, and InvestEU (EU's GGFs). For the South Korean case, we relied on desktop research and an interview with Keun Lee, a former economic advisor to the South Korean Presidency. The list of informants is summarized in [Supplementary Appendix I](#).

GGFs as financial statecraft

GGFs were first launched in 2002 by Beijing's Zhongguancun Management Committee, peaked in 2016, and numbered 1849 by 2021 (Wei et al., 2023, p. 944). Their shareholders—central, provincial, and municipal governments, along with SOEs—reflect China's decentralized approach to developmental finance. Despite regional disparities and a shortage of high-quality ventures in less developed areas, GGFs continued to grow, reaching 2126 funds with a target value of US\$1.8tr (CN¥13tr) by August 2024. Yet only 26% had reached their fundraising targets by 2021, and many sub-funds remained inactive due to a lack of credible partners (Wei et al., 2023, p. 948). Still, actual contributions in 2022 reached CN¥6.61tr, about half the total target (Pan et al., 2024, p. 13).

GGFs were created to overcome the conservatism of banks and the scarcity of private VC. Wen and Zhao (2021) have shown that banks favor SOEs and perform poorly in funding innovation. As industrial policy intensified post-2010 (Naughton, 2021), officials recognized the need for large-scale GVCs to fund high-risk start-ups. As one GGF manager explained, banks avoid 'three-nos' firms—those with no revenues, assets, or profits—especially in sectors like biopharma, leaving GGFs to fill the gap.

City-level GGFs now account for over half of all funds and 42% of realized capital (CVsource, 2023), underscoring a 'municipalist' dynamic largely overlooked in the literature (Beck & Larsen, 2025; Kennedy, 2024). Provincial funds contribute another 38%. This central-local architecture reflects a dual model: Local authorities manage the operational detail, while the central state supplies core capital. Though fewer in number, central GGFs remain structurally and strategically important to China's financial statecraft. [Table 1](#) presents those with over US\$10 billion (bn) in assets, ranked by size and sector, and a more comprehensive list of entities with over US\$1bn in assets can be found in [Supplementary Appendix II](#).

By mandate, the alignment of central-level GGF operations with the strategic plan 'Made in China 2025' (MIC25) and five-year plan is manifestly obvious in [Table 1](#). Some of the 'mega' funds focus on simple upgrading activities for SOEs or act as social cohesion funds, removed

Table 1. Central GGFs over US\$10bn and private financing.

Central GGF	Size US\$bn	Sectoral focus	Private venture capital	Size US\$bn
National Integrated Circuit Industry Investment Fund ^a	96	Semiconductors	Sequoia Capital China	55
China Structural Reform Fund	63	Large SOEs transitioning to high value-added activities in telecom, energy, and pharma	Legend Capital	48
China Railway Development Fund	54	Advanced railway equipment and infrastructure	Qiming Venture Partners	10
China Venture Capital Fund	28	Artificial intelligence, big data, cloud computing, and Internet of Things (IoT)	Matrix Partners China	10
Central Guiding Demonstration PPP Fund	25	Public utilities and environmental projects, healthcare, and elderly care.	Sinovation Ventures	7
China Innovation Fund	21	Artificial intelligence, quantum computing, blockchain, 5G and 6G communications, and cybersecurity.	Shunwei Capital	7
National Manufacturing Industry Transformation and Upgrading Fund ^b	21	Manufacturing upgrading	K2VC	1
Guoxin Central Enterprise Operation Investment Fund	21	Strengthening SOEs's in high-end equipment manufacturing, aerospace, shipbuilding, and automotive industries.	ZhenFund	1
China Internet Investment Fund	14	Digital economy and e-commerce		
Central Enterprise Rural Industry Investment Fund	14	Agritech		
China National Green Development Fund	13	Solar, wind, and hydropower, energy storage solutions, EVs		

Source: Authors' calculations based on Zero2IPO.

^a<https://sayari.com/resources/inside-chinas-largest-semiconductor-investment-fund/>.

^b<https://finance.sina.com.cn/>.

from the VC function. But most focus on investing in innovations related to 'emerging productive forces' (semiconductors, advanced manufacturing, information technology, robotics, etc.). The chosen sectors fall into the MIC25 priorities where China bets on the developmentalist logic seen elsewhere in Asia: catch up with and eventually outcompete foreign rivals in incumbent and new technologies, including through leapfrogging them *via* new technologies produced domestically at scale (Lee, 2021).

GGFs are designed to mobilize central and local state resources to leverage private VC for innovative ventures. However, their capacity to do so varies regionally: Local governments contribute around 70% in coastal areas but only 10% in rustbelt regions (Wei et al., 2023, p. 950). Even so, the largest three GGFs far surpass the scale of the top three private VC firms. Centrally owned GGFs with assets over US\$1bn total US\$484bn—four times the US\$122bn aggregate size of private VCs at the same scale (authors' calculation from Zero2IPO). Surveys also show significant state involvement in GGF governance: 31% of GGFs have fiscal or SOE supervisory officials as observers, 29% as final approvers, and 25% as members of investment committees (Wei et al., 2023, p. 953).

This underscores that private VC is not the dominant driver of innovation finance in China. The 2023–2024 initial public offering (IPO) freeze worsened this, as private VC firms—now locked into prior investments—cut funding to startups, further amplifying the role of GGFs. This stands in stark contrast to Europe's 'venture states', where the goal is to promote private VC and minimize state ownership through public–private partnerships (Klingler-Vidra, 2018).

Looking beyond the central GGFs and at the whole GGF ecosystem that includes provincial and city-level GGFs, a sectoral analysis carried out by VentureChina (a prominent Chinese venture capital research and consulting firm) has shown that GGFs as a whole do indeed conform to investing in sectors where China is known for its technical innovation boom in the past few years.

Opening the GGF black box

Some GGFs invest in innovation ventures either directly or indirectly—as limited partners (LPs) through private equity or VC firms, or as funds of funds (FoFs) channeling capital into sub-funds, typically at the provincial or city level. These sub-funds then target strategic sectors aligned with five-year plan priorities (Pan et al., 2024, p. 12). The most active GGFs are in Beijing and coastal industrial hubs, where high-tech agglomerations enable investments in 'deeptech' startups (Wei et al., 2023).

Politically, this localized investment approach helps prevent a race to the bottom among provinces and cities by reducing the need to attract firms through costly tax incentives. Most importantly, it enables local governments to build both financial resources and technical capacities for how to design a better innovation policy without resorting to fiscal transfers or soft public bank loans, a particularly sensitive issue given high levels of provincial state indebtedness. As one of the GGF officials put it: 'After becoming shareholders, as public servants we also have more opportunities and discourse the power [话语权] [i.e. the ability to speak and influence] to have in-depth exchanges with enterprises'.⁶

To better understand these forms of financial statecraft, consider the case of the National Manufacturing Industry Transformation and Upgrading Fund (NMITUF, hereinafter referred to as the Fund). This GGF was established in 2019 to answer the 14th Five-Year Plan's (2021–2025) (2021) calls for the deepening of advanced manufacturing and services ('new productive forces') amid an escalating trade war with the US and the first signs of a declining growth rate. As one official put it, 'few private companies would be capable of pumping money into business transformation amid an economic slowdown.'⁷ At the time of its establishment, the Fund was estimated at US\$21bn. Acting as a fund of funds (or 'mother fund', FoF) for central government entities, the Fund mainly focuses on innovative activities in higher value-added technology-heavy areas, such as machine tools, new materials, semi-conductors, renewable energy vehicles, quantum communication, 3D printing, and robots.⁸ To date, its largest equity investments (ranging between hundreds of millions to US\$1bn) are channeled through specific sub-funds specialized in AI, robotics, nanotech, and prevision vision equipment.⁹ The Fund is managed by China Development Bank (CDB) Capital, a professional asset management firm of the state-owned CDB, a giant state-owned bank playing a strategic role in financing MIC2025.

As Table 2 shows, nearly 60% of the fund's equity comes from central state actors—including the Ministry of Finance, state-owned financial institutions (like CDB Finance), major insurers, and low-tech SOEs, such as China Tobacco—whose resources are thus converted into risk capital *via* GGF participation.

However, not all mother fund shareholders are central level; provincial and municipal governments, VC firms, and SOEs contribute another 38.73%. While this FoF lacks private investors, others do include them (Wei et al., 2023). Overall, this reflects a model of government co-investment across multiple tiers of China's state apparatus.

Table 2. The top 10 shareholders of National Manufacturing Industry Transformation and Upgrading Fund.

Top 10 stakeholders	The percentage of the shares (rounded to 0.00%)
Ministry of Finance	15.29%
CDB Finance	13.59%
China Pacific Life Insurance	6.79%
People's Insurance Company of China (PICC)	10.19%
China Tobacco Corporation	10.19%
Beijing Economic and Technological Development Zone	6.8%
State-owned Assets Supervision and Administration Commission of Hubei Provincial People's Government	6.8%
Zhejiang manufacturing industry upgrading financial platform	6.8%
State-owned Assets Supervision and Administration Commission of Beijing Municipal People's Government	3.40%
Foshan Municipal and Guangdong Province Finance Bureau	3.40%

Source: Authors' calculations based on aiqicha.com.

Overall, the Fund has ten major sub-funds incorporated either as independent financial firms or integrated into an existing financial firm. One of the sub-funds is incorporated as Sichuan Manufacturing Collaborative Development Fund (SMCDF) with the Fund providing 29.93% of the equity (almost to the 30% regulatory limit). As illustrated in Figure 1,¹⁰ the investment portfolio of SMCDF includes a medical tech company, an internet of things firm specialized in public utility meters, and a semiconductor and an LED display firm. Importantly, SMCDF is managed by a private asset management firm (Sichuan Collaborative Revitalization Private Equity Investment Fund Management Co., Ltd.) specialized in the management of state assets. However, this firm is backed by a giant (provincial) government-owned financial holding company (Sichuan Development Holding) with US\$72.5bn under management in 2023.¹¹ Figure 2 shows that with the financial and

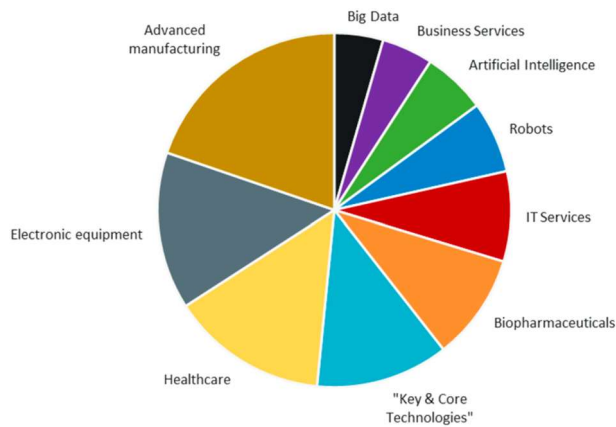


Figure 1. GGF investment by sector.
Source: García-Herrero and Schindowski (2024).

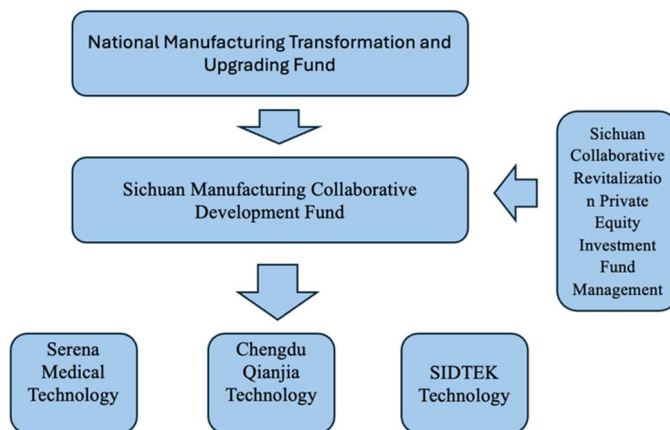


Figure 2. How 'mother' GGF connects to sub-funds.

political endorsement of this FoF, the call for setting up sub-funds can attract institutional and corporate investors to make a specific infrastructure asset investable.

China's growing reliance on FoFs reflects the need to sustain local financing in the face of economic slowdown and aligns with top-down policy signals increasingly emphasizing investment in early- and seed-stage tech projects (CVsource, 2023) while highlighting the various risks of cronyism that often plague localist dynamics in China. This should raise questions about GGFs effectiveness, an issue we turn to next.

GGF effects on the innovation economy

A rich quantitative literature in economics and economic geography has dealt with established orthodox skepticism about Chinese GGF's effectiveness. Three patterns emerge from this scholarship. First, in terms of financial performance, GGFs are not as financially agile as the country's private VC firms (Kajitani et al., 2024; Peng et al., 2024). Moreover, concerns over a 'crowding-out' effect in the VC ecosystem persist, as GGFs often struggle to attract private financial investors to their portfolios (Da Rin et al., 2006). Huang et al. (2023) have highlighted that GGF-backed firms perform well in IPO initial returns due to political capital mitigating information asymmetry, but this advantage does not translate into sustained operational performance. Such critiques underscore the limitations of state-directed VC in sustaining value creation for the shareholders.

But emphasizing the relative inferiority of GGFs on traditional performance metrics in the financial sector misses the point that by their very nature, China's neo-developmental GGFs are forms of 'grand steerage' (Naughton, 2021) of technological innovation-based growth over standard performance metrics. And indeed, in this regard, there is hard evidence that on average GGFs deliver on this mandate. Thus, econometric studies have shown that GGFs do prioritize policy-endorsed sectors, invest at earlier stages, and hold equity longer than banks, while supporting innovation in less developed regions where financial opportunities are more limited (Ge et al., 2024). On average, by 2021 the sectors which accounted for more than half of all GGF investments were in telecommunications, computers, and medical and pharmaceutical products (Wei et al., 2023, pp. 945–946), three areas where Chinese firms have risen to challenge established global firms.

Most importantly, 'with policy backed firms sponsored by GVCs having 14.9% more patent applications than those backed by non-GVCs' (Ge et al., 2024, p. 2), GGFs are also more spatially distributed than private VC funds, covering even less-developed inland areas (Beraja et al., 2024).

GGF-backed firms in policy-driven sectors may underperform in IPOs and mergers and acquisitions (M&A) compared to those that are private VC-backed, but they do play an essential role in fostering innovation and scaling it up, particularly when concentrated in high tech zones, as demonstrated by Wei et al. (2023, pp. 945–948). Still, success varies across sectors. While GGFs played a role in the electric vehicle value chain, they have been less successful at helping China catch up on advanced chips, perhaps the most geoeconomically valuable good (Fuller, 2016).

Third, some GGFs go beyond pure VC activities to support local initiatives in talent attraction, infrastructure development for innovation clusters, as well as in funding advanced technology transfer activities (Standaert & Manigart, 2018). This strategic orchestration of resources ensures that the innovation-driven economy also gets supported by financing the ‘hard’ infrastructure these innovation zones need. Studies of high-tech zones have confirmed GGFs’ transformative impact, with their equity allocations correlated to a 5.5% surge in industrial activity and significant agglomeration benefits (Xiao et al., 2024). Yet some studies have cautioned that in doing so GGFs can get trapped in infrastructure public-private partnerships (PPP) rather than foster innovation (Kajitani et al., 2024). Further challenges persist, including lower-than-expected fundraising outcomes (Wei et al., 2023), limited exit diversification (Lau, 2024), overcapacity exacerbating international trade tensions (Xiao & Yuen, 2024), and inefficiencies caused by overlapping priorities (Wei et al., 2023). Finally, in an increasingly geoeconomic world, Chinese financial statecraft intersects with American financial counter-statecraft. A recent study (Pan et al., 2024) has highlighted how the US has significantly restricted the ability of American private equity and VC firms, as well as advanced business services essential to the globalization of Chinese innovation, to operate within the Chinese market. These measures have escalated the costs of innovation intermediation, exposing the limited international brokerage capacities of GGF.

Nonetheless, there seems to be a consensus in these econometric studies that with certain qualifications, GGFs can effectively abet innovation ambitions, forming a core component of China’s distinctive model of neo-developmental financial statecraft. As the next section shows, GGFs blend central oversight with local and experimental execution of sectoral innovation priorities assumed by different levels of indicative planning.

Localizing and disciplining investment

By law, local sub-funds must invest in tech ventures within their own region, targeting sectors like semiconductors and electric vehicles through

private equity and VC firms. A key constraint is China's unique *reverse investment ratio* (RIR), which requires sub-funds to invest locally at 100–200% of the central FoF's equity commitment (Wang, 2022). However, recent reports have suggested that enforcement of the RIR has relaxed toward the lower end (China-FOF, 2023). Another requirement—the FoF-to-sub-fund investment cap—limits FoF contributions to 20–30% of a sub-fund's capital (Zhongzhengnet, 2018), though some cities now permit contributions of up to 50%, or even 60% in select cases (Luong et al., 2021). These regional adaptations ease fiscal pressure on local governments while channeling more top-down support toward early-stage, high-risk tech ventures. As the manager of a state-owned Ningbo Venture Capital Guiding Fund (NVCGF) revealed to us:

We plan to increase our fundraising efforts at the General Partner (GP) level, thereby enhancing our controlling and managerial power within the fund. On one hand, we mitigate the fiscal pressures of sub-regional levels, and on the other hand, we aim to provide funds with longer horizons.¹²

As shown in the last empirical section, the intense localization sits in great contrast to GGFs in derisking states, which are driven by demand-based investments with no hard geographical conditionalities imposed. For example, the 2023 Ningbo Administrative Measures for Government Guidance Funds requires that sub-funds invest at least 120% of the FoF's contribution into 'real economy' enterprises located in Ningbo—a reduction from the 150% threshold set in 2019 (Ministry of Finance of Ningbo, 2019). If a FoF invests CN¥1bn in a Ningbo sub-fund, the fund manager must commit at least CN¥1.2bn to local tech ventures, even if more lucrative opportunities exist elsewhere, such as in Shanghai or Hong Kong. These regulatory restrictions on where to invest and how much to invest locally seek to lock in capital resources to ease the funding difficulties of high-innovation firms in strategic sectors in that specific region where the sub-fund is established. These localization requirements did yield some results. As Ge et al. (2024) have demonstrated, there is hard evidence that Chinese GGFs are more likely to invest in such regions than private VC firms are.

A defining feature of China's GGF system is the creation of local state asset managers with a public mission—unlike derisking states where private asset managers prioritize shareholder value. Chinese regulations link manager compensation directly to local investment performance, especially under the RIR mandate. While European local GVCs lack such formal links (Testa et al., 2024), Chinese GGF managers—typically asset management firms—can earn substantial profit-sharing bonuses. For instance, Foshan's GGF rules grant managers up to 40% of excess profits for meeting a 150% RIR target, and up to 50% if they exceed 200%

(Finance Development Office of Foshan, 2019). These incentive structures foster a national and local financial technocracy tasked with managing public investment through market-style tools—but aligned with industrial policy goals. Their performance is tied not just to returns but to scale, localization, and coherence with provincial development plans, ultimately supporting national technological upgrading.

Alongside generous incentives, the GGF system also imposes penalties. Article 12 of the Ministry of Finance's (2015) guidelines prohibits GGF managers from investing in non-innovative activities like mortgages, stocks, or real estate. Central-level investors enforce discipline by threatening to withdraw equity if innovation mandates are breached. Some local GGFs gain operational autonomy by helping officials meet tech targets—absorbing losses when investments fail—a form of 'political derisking'. Yet this flexibility often conflicts with strict shareholder value enforcement. As one Shenzhen GGF manager put it:

As a state-owned venture capitalist, we bear more pressure. For example, our money does not come from ourselves, but from higher-level state-owned investors. If not handled properly, we may be held accountable by higher-level investors. This means that on one hand, we need to be cautious to avoid the loss of state-owned assets. On the other hand, we are aware of providing patient capital as advocated by the central government. As many of you know, we have filed repurchase claims against several of our funded firms, as they failed to go public within the agreed-upon time-frame stipulated in our agreements. I know it is a hard situation for them because the Chinese IPO market is getting rigid, but we are also having a hard time because we don't want to lose state money.¹³

The main downside of mandatory localization is that less developed provinces or municipalities might not have a solid technological and corporate base to provide enough investment opportunities for such GGFs (Sun & Tian, 2024), leading to the relaxation of RIR over time.¹⁴ To counter this risk, many provincial leaders have therefore scrambled in a national race to develop world-class technical universities and innovation clusters churning out talent and innovation hubs that can ensure a steady flow of quality innovation projects for GGFs to fund.¹⁵ In this regard, informants revealed the strong benefits of the alignment of GGFs and industrial cluster planning.¹⁶

Also, some of our interviews highlight tensions between government officials, state VC, and private VC, including restrictive localization requirements and conflicting mandates. One GGF informant likened this to 'dancing with chains—and we must dance well'.¹⁷ As another one put it:

It is in our mutual interest that they (local government, *authors' note*) refrain from interfering with our investment decision-making and exit

strategies. As an experienced finance team, we are committed to adhering to market principles rather than administrative directives. For government officials, this approach minimizes the risk of being held accountable for any potential state capital losses resulting from unsuccessful investments.¹⁸

Localization carries the risk that local authorities may steer GGF capital toward low-risk, late-stage firms in incumbent sectors rather than early-stage ventures. However, recent closures and equity withdrawals by central FoFs have curbed this trend, reinforcing central oversight (Luong et al., 2021, p. 59). With national entities steering innovation finance, the threat of disciplinary action remains credible.

Thus far, the paper has shown that China's GGFs stand out for converting state assets into risk capital through centrally coordinated, geographically targeted investment aligned with sector-specific goals across a multi-level state. But how unique is this model? The next section compares it with GGFs in a derisking state (EU) and a liberal neo-developmental state (South Korea).

Chinese GGFs in comparative perspective

This section compares the results of the previous discussion on China with South Korea-based GGFs (South Korea Venture Investment Corporation or KVIC) (Chung & Kang, 2018; Klingler-Vidra & Pardo, 2020) and EU-based ones (EIF, Invest EU, European Investment Council, national and regional GGFs across the EU member states) (Mertens & Thiemann, 2023; Mocanu & Thiemann, 2024) not just in terms of comparative size but also across the variables posited as key by the theoretical framework: mobilization of both fiscal resources and state-owned enterprises, sector-specificity in their technical innovation mandates, extent of coordination with national upgrading strategies and the type of conditionalities imposed on them by the state. Overall, there are some similarities between all these financial vehicles. All are based on an acknowledgement of market failure and the principle that publicly-owned financial entities should take a lead in innovation financing. All provide loans to businesses directly as well as indirectly through other financial intermediaries to provide government VC funding for innovative but riskier activities. All delegate management of the fund to financial actors and combine public and private resources together. Finally, all operate under executive authority at the state level, and all are owned overwhelmingly by government entities, be they institutions or state-owned financial firms (see Table 3).

However, as hypothesized, the Chinese GGFs remain sharply distinctive from South Korean or European GVCs, a distinctiveness which reflects

Table 3. Ownership comparison of state investment funds.

Entity	Country/ region	Legal form	Sole/principal shareholder(s)	Notes on ownership and governance
Chinese Government-Guided Funds (GGFs)	China	Investment partnerships/LP structures	Local/provincial governments as anchor LPs; SOEs, state-owned banks, insurance companies, and state-owned managers of state assets (80%); private capital (20%)	Public capital typically dominates but is often mixed with private LPs; government sets mandates but does not always manage directly.
European Investment Fund (EIF)	European Union	Public-private partnership (EU legal entity)	EIB (59.8%), European Commission (29.7%), Financial institutions, such as national development banks and private banks (10.5%)	Minority shareholders include national promotional and private banks; governance through General Meeting and Board.
European Innovation Council (EIC) Fund	European Union	Reserved Alternative Investment Fund (RAIF) managed by the EU's European Investment Bank	European Commission (100%)	Managed by external entity AIFM; advised by EIB; targets equity stakes in EIC Accelerator portfolio firms.
Korea Venture Investment Corp. (KVIC)	South Korea	Government-owned corporation	South Korean government, operations are overseen and funded by the Ministry of SMEs and Startups (MSS) (100%)	Acts as fund-of-funds; allocates capital to private VCs via public LP commitments; directly accountable to government.

Sources: For the European Investment Fund (EIF), data was sourced primarily from the EIF's official Register of Shareholders (https://www.eif.org/who_we_are/shareholder/register.htm) and its Annual Report of 2023 (https://www.eif.org/news_centre/publications/eif-annual-report-2023.htm). Information on the European Innovation Council (EIC) Fund is based on documentation from the European Commission's official EIC Fund page (https://eic.ec.europa.eu/eic-fund_en#:~:text=EIC%20Fund) and investment guidelines provided by the European Investment Bank (https://eic.ec.europa.eu/system/files/2022-12/220301%20EIC%20Investment%20Guidelines%20-%20Horizon%20Europe%20March%202022%20FINAL%20%281%29_0.pdf). Details on Chinese GGFs are drawn from 2024 Active Institutional LP Profile Report (http://www.chinahightech.com/chuangye/2025-01/16/content_286678.html). For South Korea Venture Investment Corp. KVIC information comes from KVIC's official website (<https://www.kvic.or.kr/en/>) and the Ministry of SMEs and Startups (<https://www.mss.go.kr/site/eng/main.do>).

the structural characteristics of the China's dirigiste neo-developmental state. First off, even by conservative estimates, GGFs are much larger. Table 4 below highlights this finding with triangulated estimates of public innovation investment through state-backed VC and industrial policy

Table 4. Triangulated estimates of public innovation investment in South Korea, EU, and China.

Jurisdiction	Public innovation finance (US\$ bn)	% of GDP	% of Financial sector assets	Notes
South Korea	8.2	0.48%	0.18%	Based on GDP 1700 and assets 4500
European Union	155.66	0.84%	0.17%	Based on GDP 18,600 and assets 90,000
China	489.6	2.75%	0.72%	Based on GDP 17,800 and assets 68,000

Source: See [Supplementary Appendix III](#).

vehicles in the EU, China, and South Korea. This is the case whether we looked at the total value of financial commitments, its share of financial sector assets, and share of GDP. Interestingly, while EU and South Korea have GVCs worth a similar share of total financial sector assets, in the EU, they are almost twice as large a share of GDP. The estimates are harmonized in US\$ and cover the period up to 2023 (See [Supplementary Appendix III](#) for sources and calculations).

While bigger, GGFs are less patient to the firms in their portfolio than European and South Korean public funds. While 10–12 years is a standard timeframe in Europe and South Korea, in China, it is much shorter. As several interviewees noted, the typical 6–7-year horizon of GGFs often falls short, as many tech firms require 12–15 years to get mature. This shortened duration is partly driven by the tenure of government officials overseeing the funds (normally stay at one position for 5 years and two terms maximum). These officials are often eager to demonstrate successful outcomes, such as profitable exits or measurable policy impacts, before they transition to new roles. As a result, they wish to promote safely and land safely, denting the main intent of GVC: patient capital for high-risk ventures.¹⁹

Second, in light of the mobilization of both fiscal resources and state-owned enterprises, Chinese GGFs rely less on fiscal authorities for capitalization and draw overwhelmingly on equity from state-owned firms, state owned financial institutions (bank and non-bank alike) as well as the entities of local governments and municipalities, thus multiplying their capital pool. EU and South Korean GVCs rely almost exclusively on fiscal resources and are, therefore, less ample given budgetary constraints. While private venture investors have contributed over 50% of funding for both South Korean (Shin, 2014, p. 36) and European GVC initiatives at the national level (Testa et al., 2024), their contribution to Chinese GGFs has remained significantly lower, typically in the low teens.²⁰

Third, in terms of innovation mandates, GGFs have stronger mandates for high value-added technological innovation. Chinese projects supported

by GGFs are managed largely by state-owned financial entities incentivized to maximize investment in the chosen sectors targeted by the planners. This creates important advantages for China's financial statecraft: While management compensation in the European and South Korean models is shaped by shareholder value and credit ratings, in China the focus is more on technology innovation targets. Until the recent creation of the modest European Innovation Council Fund (EICF) in 2020, the EIF lacked a mandate to establish sector-specific sub-funds aligned with strategic innovation goals set by its principal, the EC. Although a more statecraft-minded Commission in 2017 pushed the EIF to support innovation and shield EU start-ups from foreign acquisition (Testa et al., 2024), no specific technological mandate was defined.

As shown in Table 5 below, the EIF remains a generalist allocator of the EIF's sector-specific funds at exit from year 2024 to 2029. 'Generalist' category topped the list in terms of the number of investees accounting for 41.3% of the investment share, followed by 31.4% share on digitalization and 10.5% on cleantech. In line with the liberal neo-developmental regime, South Korea's FoF (KFoF) stands out in terms of its investment category of 'Culture/Media/Entertainment', accounting for 13% of the number of portfolio companies in the year 2021. Hence, aside from the focus on digitalization, which is part of EU and South Korean government strategy, we see that neither EIF nor KFoF has a clear innovation mandate with diverse sectoral targets like China's GGFs do. The EU€3.5bn EICF is a notable exception: the first EU fund to directly take equity stakes in high-risk, high-impact start-ups across stages—from research to scale-up—designed as a true 'unicorn factory' (Mocanu & Thiemann, 2024, p. 445).

Moreover, EIB-managed InvestEU is of no exception. On the one hand, it has a demonstrated track record in focusing its investments on high value added sectors, with its initial investments including advanced geothermal energy systems, hydrogen-powered steel production, or cutting-edge battery manufacturing systems.²¹ Yet in response to member states' demand it did so in a diluted way, alongside a 'Christmas tree' of missions unrelated to innovation, such as general countercyclical lending

Table 5. GVC investments by sector.

Sector	EIF investment share	KFoF investment share
Digitalization	31.4% ICT	31.2%
Cleantech/energy/ESG	10.5%	NA
Bio/healthcare	12.3%	18.2%
Generalist	41.3%	Not applicable
Consumer/distribution	1.4%	18.9%
Culture/media/entertainment	Not applicable	13%
Industrial/manufacturing	3.1%	12.2%
Other	Not applicable	6.6%

Source: See Supplementary Appendix IV.

to SMEs, job growth, digital inclusion, skill upgrading, health and water infrastructure etc., thus trying to be both a venture capitalist and a supplier of social cohesion.²² The situation is not different at the national and sub-national GVC level. Here, Testa et al. (2024, p. 27) have found that out of 369 European GVC initiatives screened, 81% did not have a specific industry focus, and only 19% (almost all in Northern Europe and France) had a mandate to invest in a specific sector, most typically biotechnology, digital technologies, energy, health-related technologies. Unlike in the case of China, the actual impact on technological innovation of these funds is uncertain.

However, if we only compare between South Korea and Europe, we argue that the liberal neo-developmental state in South Korea brings a stronger sector orientation relative to the EU. KVIC acts as a meta-GP and strategic coordinator of underlying funds under different thematic umbrellas. These include KFoFs (core fund-of-funds), Foreign VC funds (global partnerships in local ventures), policy funds (e.g. green, youth, digital), and angel fund. All these VC structures are operated under KVIC's broader mandate, making KVIC one of Asia's most diversified public venture investors. KFoF, for example, has a dozen ministries as its LPs and invests in VC funds operated by private VC firms. These GVCs exhibit a more explicit focus on high-value added sectors (Klingler-Vidra & Pardo, 2020, pp. 314–315), a reflection of the developmentalist mindset captured by Thurbon (2016) and aligned with the financial statecraft perspective adopted in this paper. Its investment committee includes both government and private sector experts; it co-sets sectoral priorities with ministries; and it uses tools like anchor investments and matched funding to shape fund behavior. However, it does not discipline the private sector through Chinese-style reverse investment ratios nor ban lists for non-core investments.

The mandates of GVCs not only vary in the extent of technology focus, but also in the target of ventures of different stages. This public innovation finance increasingly targets small firms, especially start-ups, reflecting a broader shift toward entrepreneur-focused funding in both derisking and liberal neo-developmental states (Klingler-Vidra & Pardo, 2020; Testa et al., 2024). In China, however, beneficiaries range from start-ups to midcaps and even large infrastructure projects, as long as they align with clearly defined industrial goals in the Five-Year Plan. In Europe, the post-2020 push for strategic autonomy has led the EU to scale up support—particularly through the EICF—for deeptech start-ups, aiming to shield them from foreign acquisition (Mocanu & Thiemann, 2024, p. 445).

Fourth, on the face of it, coordination and localization in GVCs can be seen as a generic feature of such funds everywhere. For example, the EIF coordinates with regional and national sub-funds out of concerns

for territorial cohesion, usually in partnership with local governments.²³ These are VC, private equity (PE), and mezzanine funds with deep local market knowledge or co-investment vehicles alongside angel investors, investment funds, family offices, and institutional investors into SMEs and small mid-caps at various growth stages.²⁴ Unlike the more rigid coordination in China's GGFs, which rely solely on equity as a policy instrument that ties multi-state levels together through the investment chains, European GVC agencies adopt a more flexible approach—combining equity with grants, loans, and guarantees to leverage its funds through financial intermediaries. Moreover, the relatively small size of European GVCs (the largest at EU€1bn) and their negotiation with private investors often result in a concentration of investment in established financial and innovation hubs, as private firms typically resist geographic conditionalities.²⁵ A survey of all European GGFs in 11 key EU member states found that only the regional GGFs have local investment targets, although in practice, the amounts of contributed capital required of their private VC managers to invest locally does not exceed 60% (Testa et al., 2024, p. 24). In China, the sub-funds are not just much bigger, but their asset managers are subject to formal and stricter geographical conditionality, as shown in our analysis. No such localized dynamics exists in the case of the South Korean GVCs as it operates in a unitary rather than decentralized institutional structure. As such, it does not embed spatial mandates—such as requiring capital to remain in a specific province—seen in many local-level GGFs.

Discussion and conclusions

This study invites scholars to rethink financial statecraft in an era where technological sovereignty and resilience are paramount. China's GGFs stand as a powerful 'financial statecraft' counter-narrative to derisking state recipes. Far from replicating conventional models of innovation finance, these GGFs embody a bespoke governance framework that integrates state control with market adaptability, marking a transformative approach to fostering ascendancy. In so doing, GGFs do not represent a seamless, top-down imposition of state priorities on a compliant private sector. Instead, they embody an adaptive dirigisme—a form of guidance or steerage—shaping entrepreneurial firms' motivations and resources through influence rather than direct commands.

Our findings empirically both bolster and go beyond the expectations generated by the financial statecraft framework we advanced at the beginning of the paper. Table 6 synthesizes the core focal points generated by the financial statecraft view and, with italics, the surprising elements that emerged from the empirical analysis only. In terms of the

Table 6. Derisking state, liberal neo-developmental state, and dirigiste neo-developmental state.

	Derisking state (EU)	Liberal neo-developmental state (South Korea)	Dirigiste neo-developmental state (China)
Specific technology innovation mandate	Weak, with growth and cohesion focus; <i>post-2020 stronger technology innovation emerged in the small-scale EICF.</i>	Moderate, with growth and broad knowledge economy orientation for market actors.	Strong, with specific sectoral innovation targets embedded in national and provincial indicative plans.
What state resources are converted into risk capital	Strictly fiscal resources of executive branch actors at the EU, national, and regional levels.	Strictly fiscal resources of executive branch actors at the national level.	State assets from SOEs, state-owned banks, state-owned insurance companies, state asset management firms, layered on top of fiscal resources of executive branch actors national level.
Geographic conditionality	No	No	Yes, with reverse investment ratio and <i>equity withdrawal for mandate non-fulfillment.</i>
Extent of institutional coordination	Medium (inter-agency coordination in a network state across intergovernmental and supranational levels)	Medium (inter-agency coordination in a national network state)	High (5-year macro-planning in a decentralised state)

comparative analysis with shadow cases, we found that four defining characteristics set Chinese GGFs apart. First, in China, the state remains firmly in control, aligning profit-driven investment with national strategic priorities through indicative planning mechanisms. It enforces discipline among sub-funds by banning non-core innovation activities and threatening to withdraw central-level fund-of-fund equity if mandates are not fulfilled. In contrast, the European case has evolved more dynamically than expected—moving closer to the South Korean model—with elements of neo-developmental financial statecraft emerging experimentally after 2020.

Second, when doing VC investing, only Chinese GGFs mobilize state assets from SOEs, state-owned insurance companies, or state asset management firms on top of the fiscal resources of executive branch actors at national level. In this way, only China converts ‘stale’ state assets into patient finance for productivity-enhancing technological innovations, an option available but not exercised in both Europe and South Korea, where the state still owns large parts of critical sectors like energy and insurance.

Third, only GGFs have a strict geographical diversification mandate, fostering local innovation ecosystems while maintaining central oversight.

The RIR increasing localization effects in both innovation and the formation of a GGF management technocracy appear to be a particularly innovative and distinctive feature of Chinese GGFs. This feature addresses regional disparities and enables less developed areas to develop innovation hubs while fostering hypergrowth in already developed ones. However, these mandates also reveal challenges in aligning local and national objectives, warranting further scrutiny.

Finally, China's GGFs exhibit a far stronger coordination mechanism, tightly linking central-level indicative planning with localized experimental execution across the investment chain of FoFs. In contrast, South Korea's KVIC has also managed a series of underlying funds represented by and mostly dominantly by KFoF, rendering a network state of different thematic focuses. However, these underlying funds are neither geographically nor sectorally bounded, and their largest financiers remain private investors. The coordination structure in the EU's GVC ecosystem is even looser. Although there is a strong rhetorical push at the EU level to expand innovation frontiers and foster 'European unicorns', GVC agencies across member states often pursue divergent national objectives, such as promoting economic growth or reducing regional disparities (Testa et al., 2024).

This study enriches the comparative political economy of industrial policy and innovation finance by contrasting China's dirigiste model with global counterparts. Chinese GGFs' scale, sectoral focus, and integration into centralized strategies surpass the capabilities of other frameworks, highlighting the diversity of approaches to innovation finance. The paper also makes the case for the concept of 'financial statecraft', positioning GGFs as tools that not only leverage state assets but steer the direction of market activities.

To conclude, GGFs provide an alternative model for innovation policy experts, demonstrating how state stewardship can bridge equity gaps in innovation finance without falling prey to accusations of 'corporate welfare' (Bulfone et al., 2023). However, given China's particular political economy, replicating this approach requires careful adaptation to political, fiscal, and institutional contexts. Future research should examine the long-term sustainability of GGFs and their balance between profitability and policy-driven goals. Comparative studies could also explore how similar mechanisms are innovated in other regions and explore the different contextual underpinnings that make 'varieties of financial statecraft' possible, thus offering broader insights into state-led financial governance, the megatrend of an increasingly geoeconomic world. Finally, future scholarship is left with ample opportunities to deepen the discussion of the implications of these findings for the broader literature on financing innovation under state capitalism using full-fledged comparative approaches.

Notes

1. According to CVsource, the target capital of GGFs amounts to CN¥13 trillion (tr) (or US\$1.78tr at the 2024 exchange rate), accessed at <https://www.fxbaogao.com/detail/4163438>. As we show below, conservative estimates place it at less than half that.
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3. Xi Jinping, 'Secure a decisive Victory in Building a Moderately Prosperous Society', speech delivered at the 19th National Congress of the Communist Party of China, 18 October 2017, http://news.xin-huanet.com/english/special/2017-11/03/c_136725942.htm (accessed Dec 18, 2024).
4. Central Committee of the Communist Party of China (CPC) and the State Council issued, *Outline of the National Innovation-Driven Development Strategy*, [中共中央 国务院印发《国家创新驱动发展战略纲要》], *Xinhua News Agency*, May 19, 2016, translation available at <https://cset.georgetown.edu/publication/outline-of-the-national-innovation-driven-development-strategy/>.
5. Given its smaller size and unitary state structure, Korea is not a relevant comparator in this case.
6. Interview 8.
7. Tian Yun, vice director of the Beijing Economic Operation Association, in <https://www.globaltimes.cn/content/1170551.shtml>.
8. <https://aiqicha.baidu.com/s?q=%E5%9B%BD%E5%AE%B6%E5%88%B6%E9%80%A0%E4%B8%9A%E8%BD%AC%E5%9E%8B%E5%8D%87%E7%BA%A7%E5%9F%BA%E9%87%91&t=0>.
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10. Based on <https://www.tianyancha.com/company/5130876813>.
11. https://www.sdhholding.com/about_us.
12. Interview 3.
13. Interview 7.
14. Guo Libo idem.
15. Panel with academic experts on innovation at the University of the Chinese Academy of Sciences (2024).
16. Interview 2.
17. Interview 9.
18. Interview 13.
19. Interviews 2, 3, 8, 14, 15, 2024.
20. This information is validated by all the China informants.
21. Page 27 of https://commission.europa.eu/about/departments-and-executive-agencies/economic-and-financial-affairs/evaluation-reports-economic-and-financial-affairs-policies-and-spending-activities/interim-evaluation-investeu-programme_en.
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